BUILDING A SWITCH AND ROUTER NETWORK

Introduction

This lab focuses on building a functional network using a router, switch, and two end devices. The objective is to review and apply foundational Cisco IOS configuration commands by setting up the network topology, assigning IP addressing (both IPv4 and IPv6), and verifying end-to-end connectivity. By configuring and troubleshooting real network hardware, this exercise reinforces key networking concepts such as interface management, default gateway configuration, password security, and routing table verification.

Addressing Table

Device	**Interface**	**IP Address / Prefix**	**Default Gateway**
R1	G0/0/0	192.168.0.1 /24	N/A
R1	*G0/0/0*	2001:db8:acad::1/64	*N/A*
R1	*G0/0/0*	fe80::1	*N/A*
R1	G0/0/1	192.168.1.1 /24	N/A
R1	*G0/0/1*	200:db8:acad:1::1/64	*N/A*
R1	*G0/0/1*	fe80::1	*N/A*
S1	VLAN 1	192.168.1.2 /24	192.168.1.1
PC-A	NIC	192.168.1.3 /24	192.168.1.1
PC-A	*NIC*	2001:db8:acad:1::3/64	fe80::1
PC-B	NIC	192.168.0.3 /24	192.168.0.1
PC-B	*NIC*	2001:db8:acad::3/64	fe80::1

Objectives

Part 1: Set Up the Topology and Initialize Devices

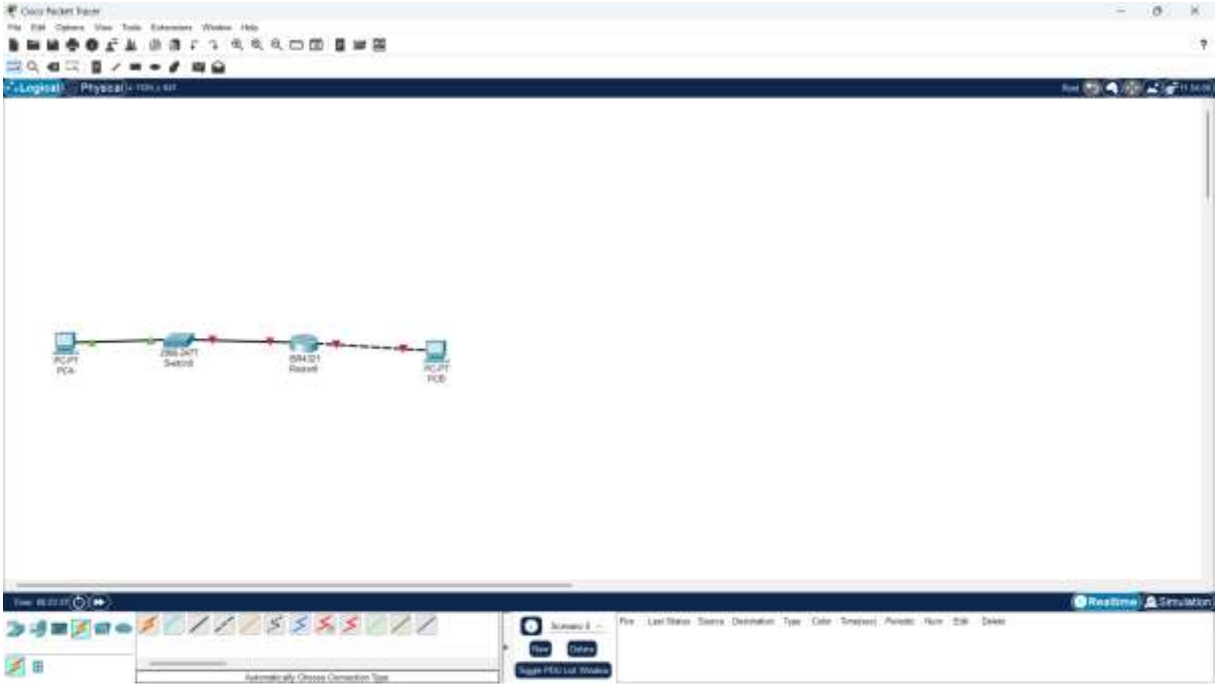
Part 2: Configure Devices and Verify Connectivity

Part 1: Set Up the Topology and Initialize Devices

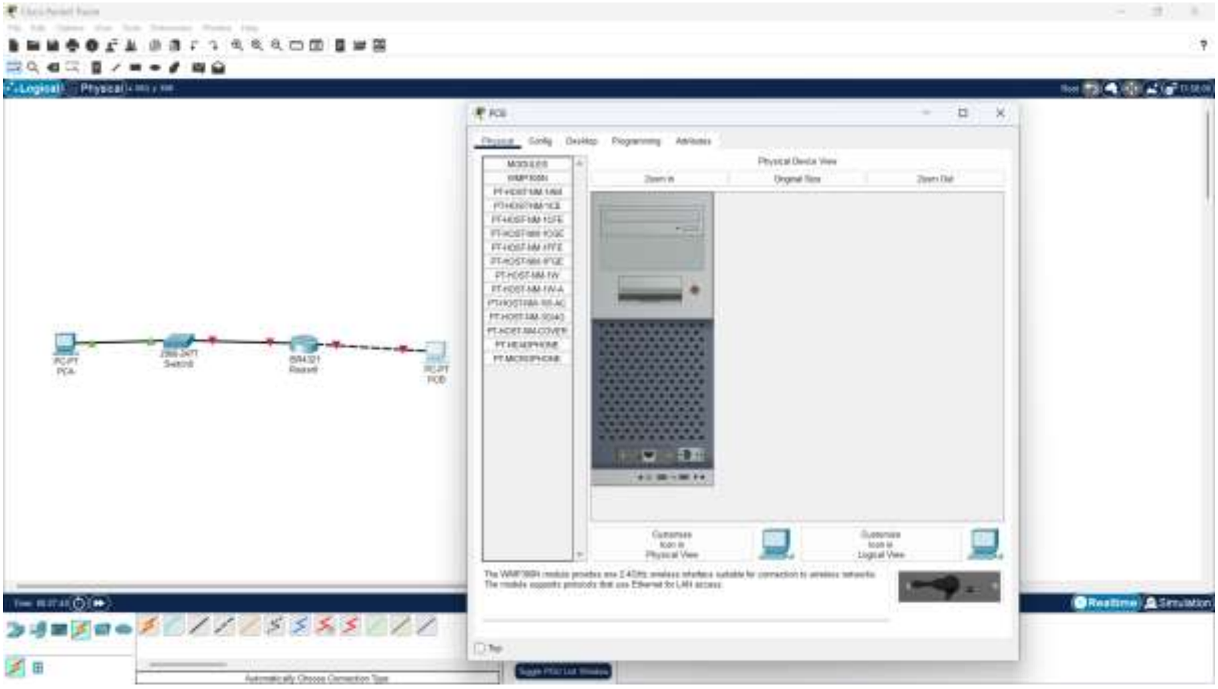
Practical

Step 1: Cabling the Network as shown in the Topology

I started by cabling the Devices as shown in the topology using Router ISR 4321, Switch 2960, PCA and PCB as shown below:



I then powered all the devices one by one

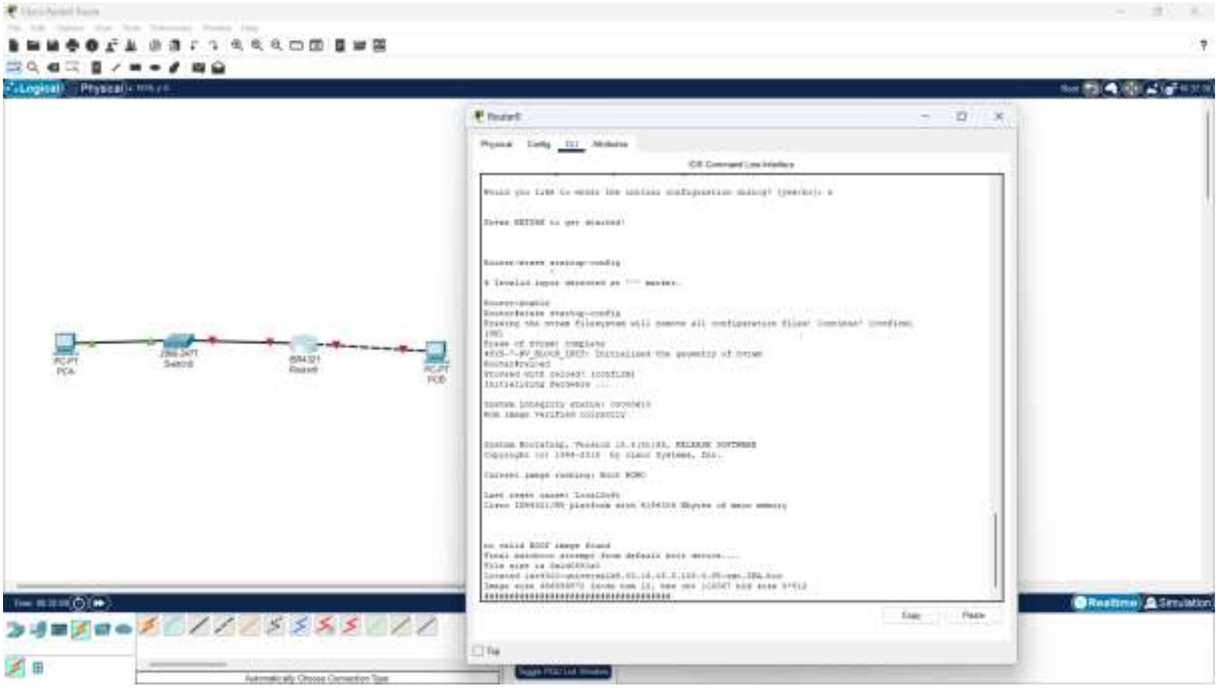


Step 2: Initialize and reload the router and switch

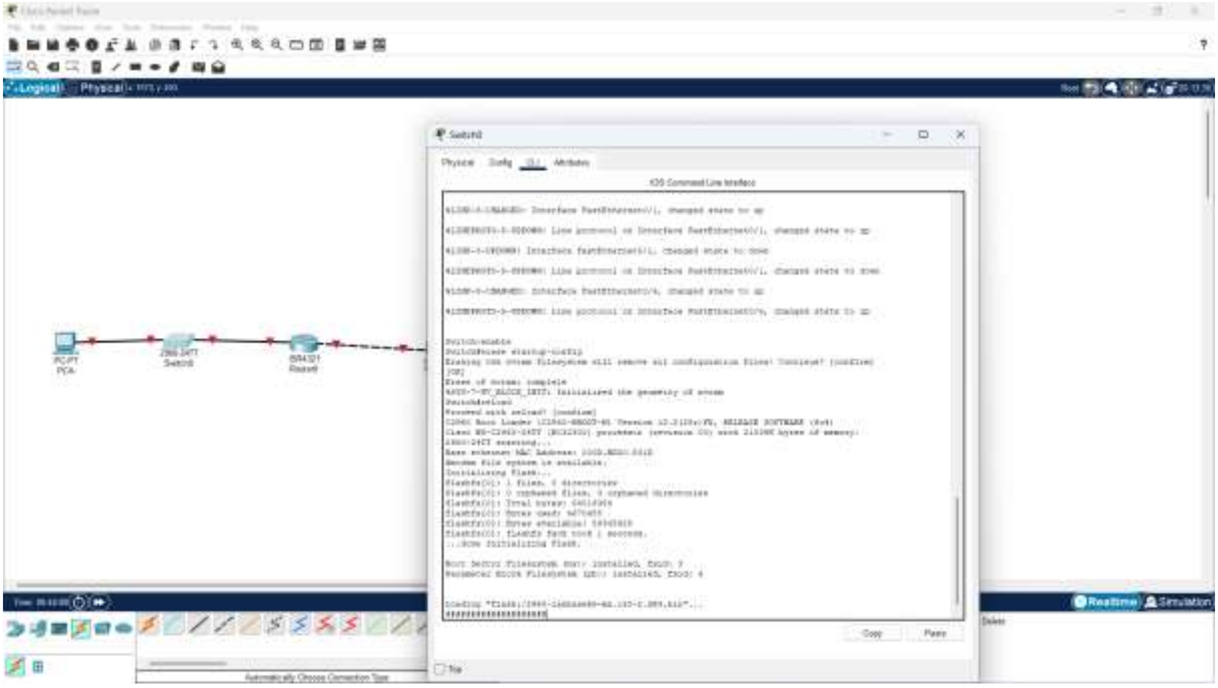
I then went ahead to Initialize and reload the router and switch.

I started by erasing the previous configurations on the router using the following steps:

1. I entered the CLI interface of the router, prevented it from using the previous configurations by typing “n” when whether I wanted to use the initial configurations.
2. Enabled the router using the “enable” command.
3. I then erased previous configurations using the “erase startup-config” command.
4. Reloaded the router using the ”reload” command.



I did the same to the switch using the exact steps and commands on the CLI interface of the switch:



PART2: Configuring Devices and Verifying Connectivity

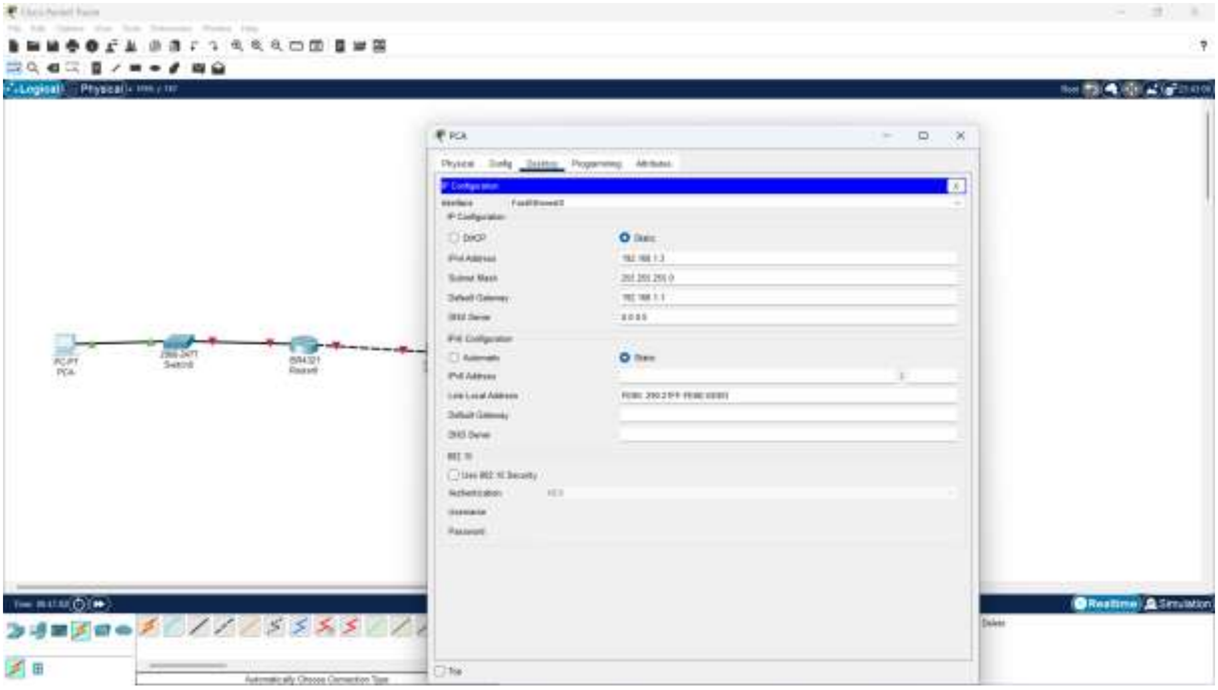
Step 1: Assigning static IP Addresses to the PC interfaces

For PCA, I assigned:

Ip 192.168.1.3

Subnet 255.255.255.0

Default Gateway: 192.168.1.1

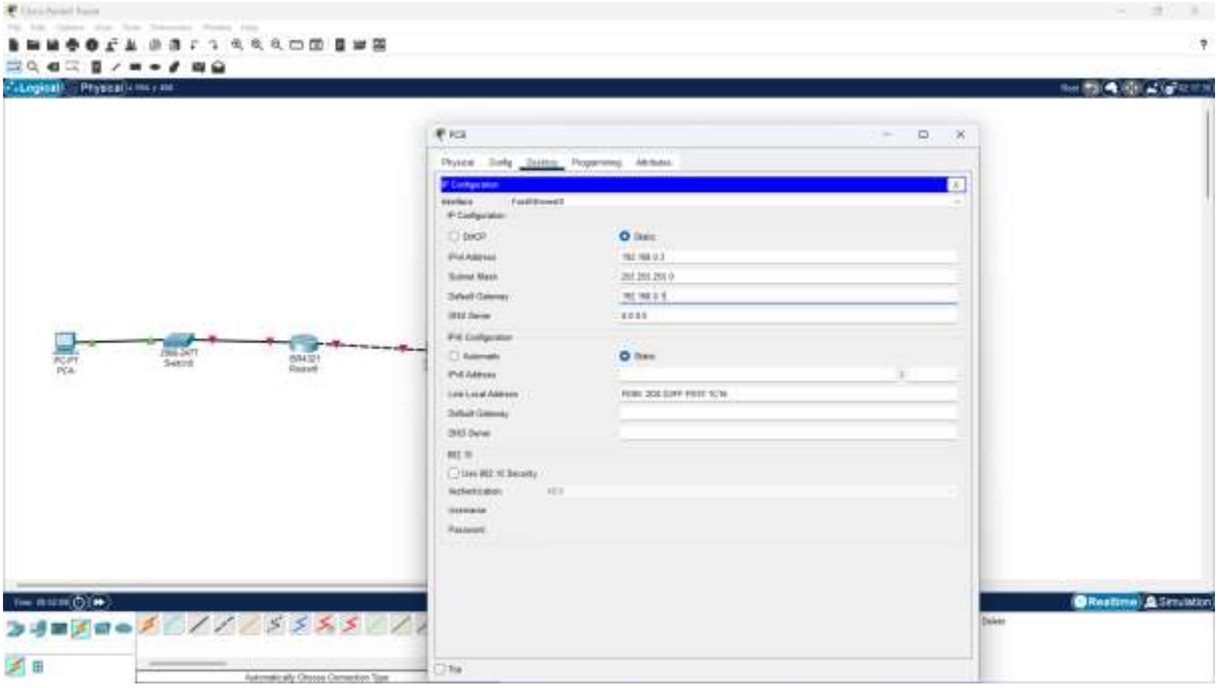


For PCB I Assigned:

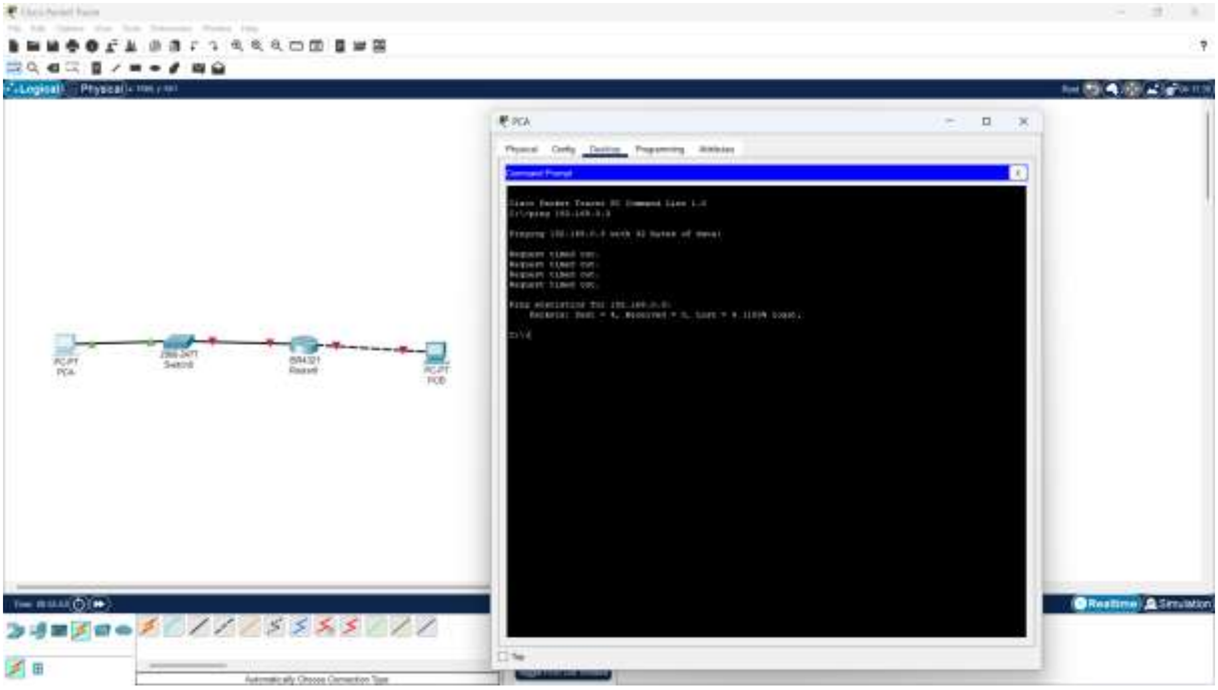
Ip:192.168.0.3

Subnet:255.255.255.0

Default Gateway:192.168.0.1



I then verified connectivity by pinging PCB from a command prompt window in PCA. The ping was unsuccessful as shown:



This shows that the connection is not yet successful, and some configuration need to be done.

I started by checking whether windows firewall was on and it was off.

I proceeded to the next level of configuring the router.

Step 2: Configuring the Router.

a) I consoled into the router CLI and enabled the EXEC mode:

Router> **enable**

b) I then entered the router configuration mode:
Router# **config terminal**

c) I then assigned a device name to the router
Router(config)# **hostname R1**

d) I then disabled DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.:
R1(config)# **no ip domain lookup**

e) Assign class as privileged EXEC encrypted password
R1(config)# **enable secret class**

f) Assigned cisco as the console password and enabled login
R1(config)# line console 0

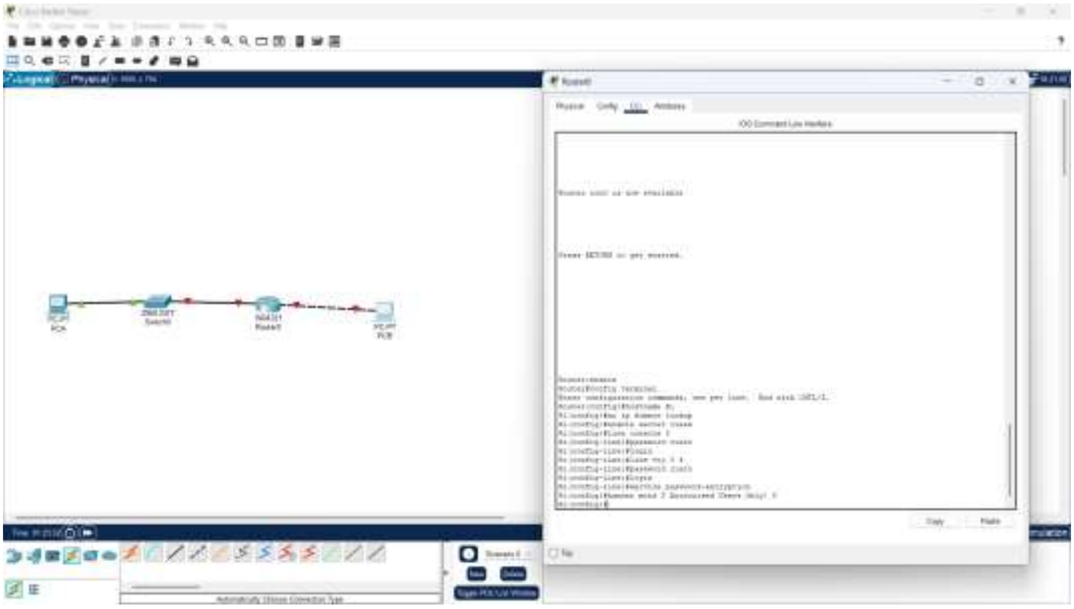
R1(config-line)# password cisco

R1(config-line)# login

g) Assign **cisco** as the VTY password and enable login.
```
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
```

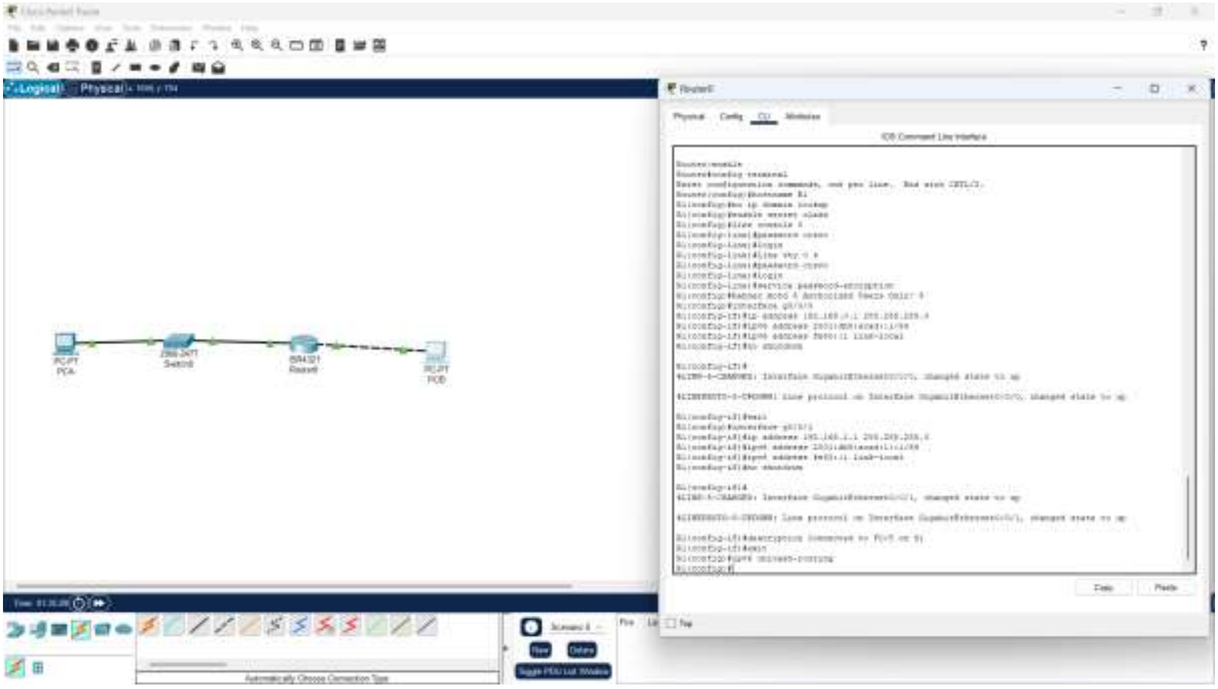
h) Encrypt the plaintext passwords.
```
R1(config)# service password-encryption
```

i) Created a banner that warns anyone accessing the device that unauthorized access is prohibited.
R1(config)# **banner motd $ Authorized Users Only! $**



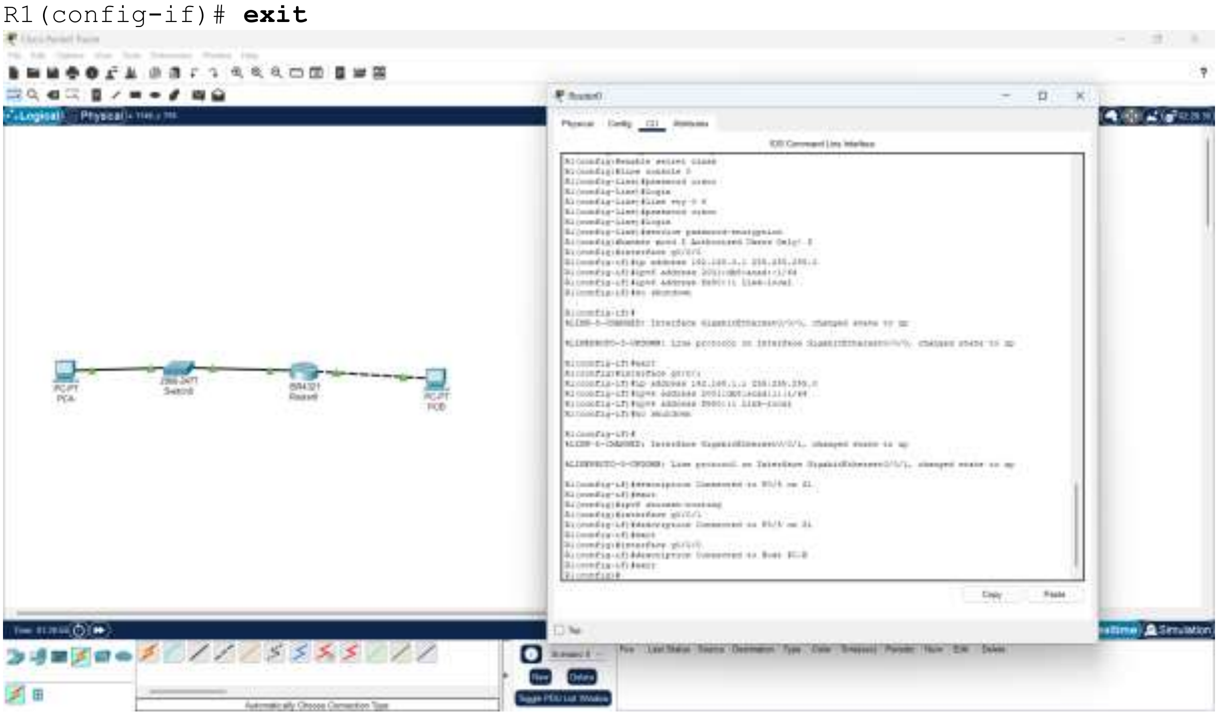
j) Configured and activated both ipv4 and ipv6 interfaces on the router.
```
R1(config)# interface g0/0/0
R1(config-if)# ip address 192.168.0.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad::1/64
R1(config-if)# ipv6 address FE80::1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)# interface g0/0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# ipv6 address fe80::1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
```



k) Configure an interface description for each interface indicating which device is connected to it.

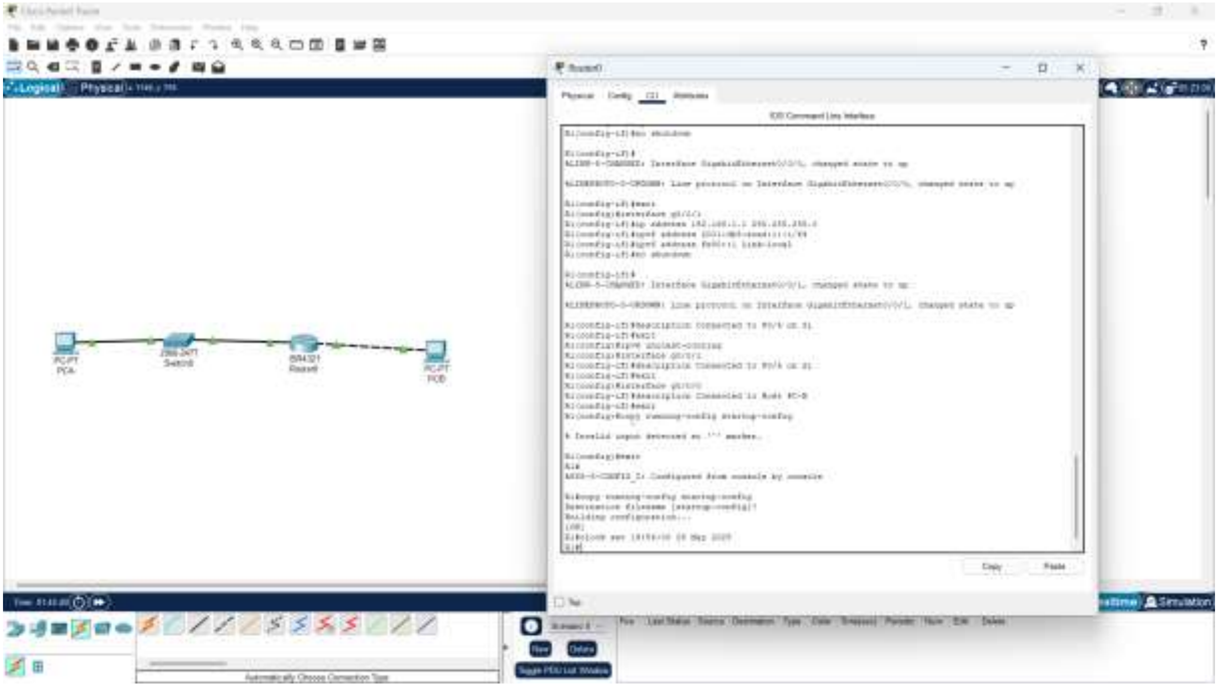
```
R1(config)# interface g0/0/1
R1(config-if)# description Connected to F0/5 on S1
R1(config-if)# exit
R1(config)# interface g0/0/0
R1(config-if)# description Connected to Host PC-B
R1(config-if)# exit
```



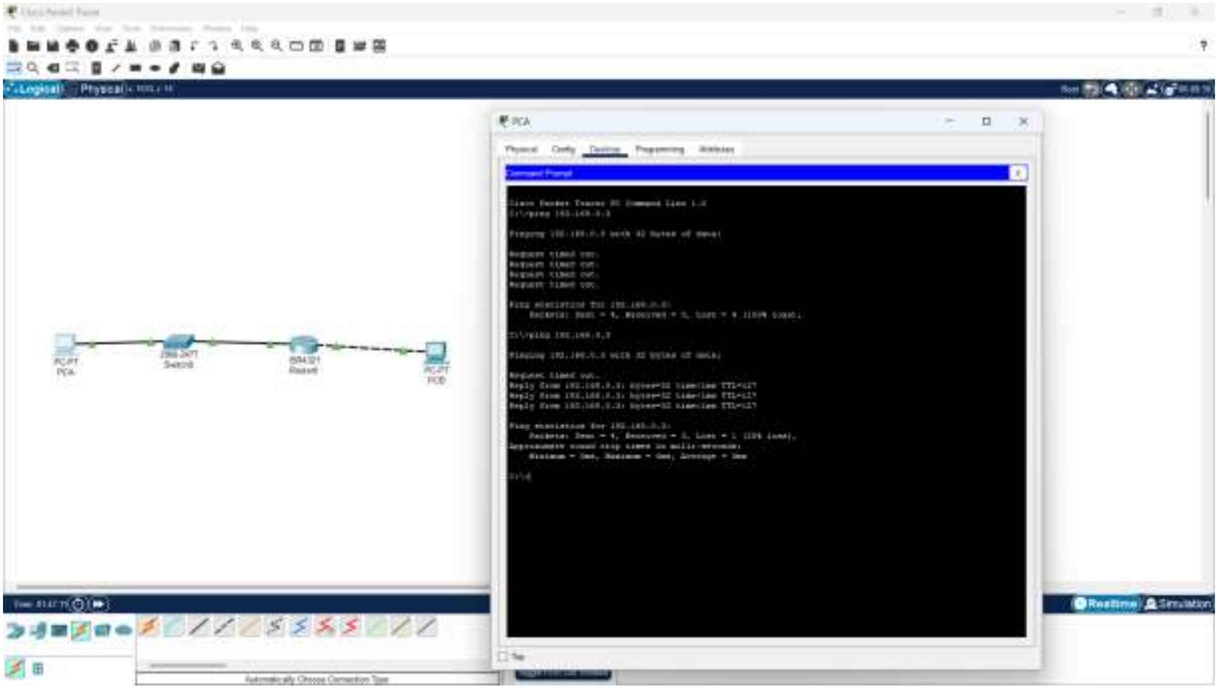
l) To enable IPv6 routing, enter the command ipv6 unicast-routing.
`R1(config)#` **`ipv6 unicast-routing`**

m) Save the running configuration to the startup configuration file.
`R1(config)#` **`exit`**
`R1#` **`copy running-config startup-config`**

n) Set the clock on the router.
`R1#` **`clock set 19:54:00 29 May 2025`**



o) I then pinged PCB from command prompt window on PCA. The ping was now successful since the configuration was now successful.



STEP3: Configuring the Switch

In this step, I configured the switch's hostname, the VLAN 1 interface and its default gateway.

a. I Consoled into the switch and enable privileged EXEC mode.

Switch> **enable**

b. Enter configuration mode.

Switch# **config terminal**

c. Assign a device name to the switch.

Switch(config)# **hostname S1**

d. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

S1(config)# **no ip domain-lookup**

e. Configure and activate the VLAN interface on the switch S1.

S1(config)# **interface vlan 1**

S1(config-if)# **ip address 192.168.1.2 255.255.255.0**

S1(config-if)# **no shutdown**

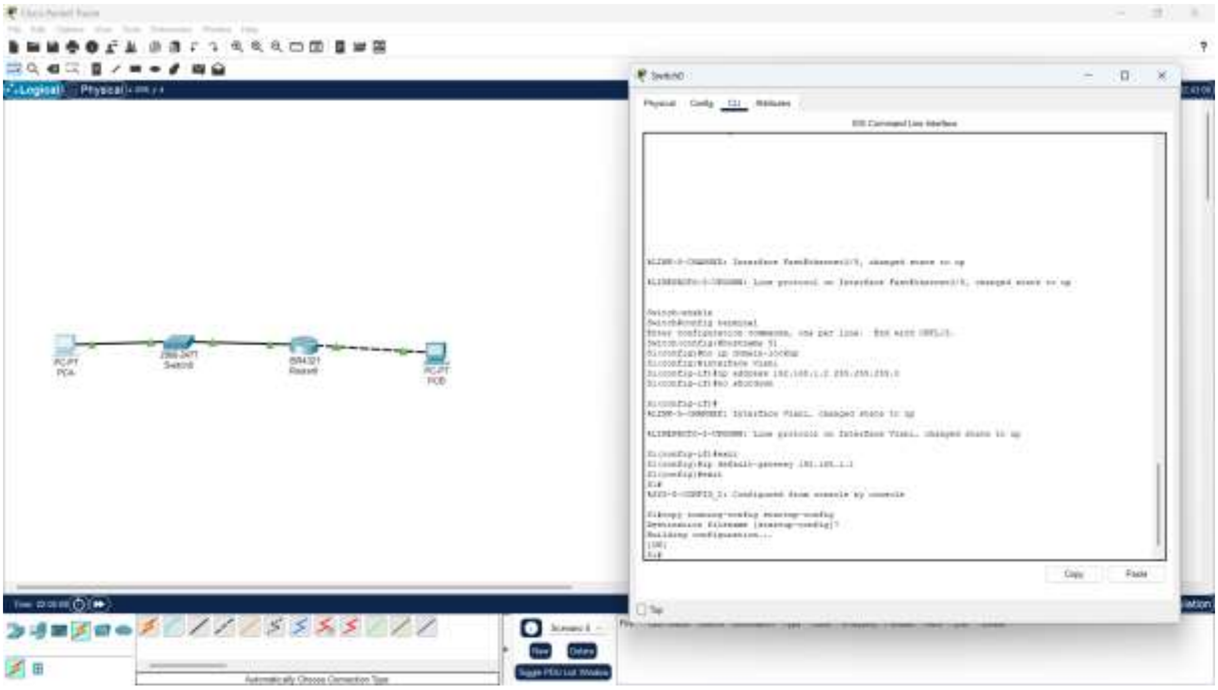
S1(config-if)# **exit**

f. Configure the default gateway for the switch S1**.**

S1(config)# **ip default-gateway 192.168.1.1**

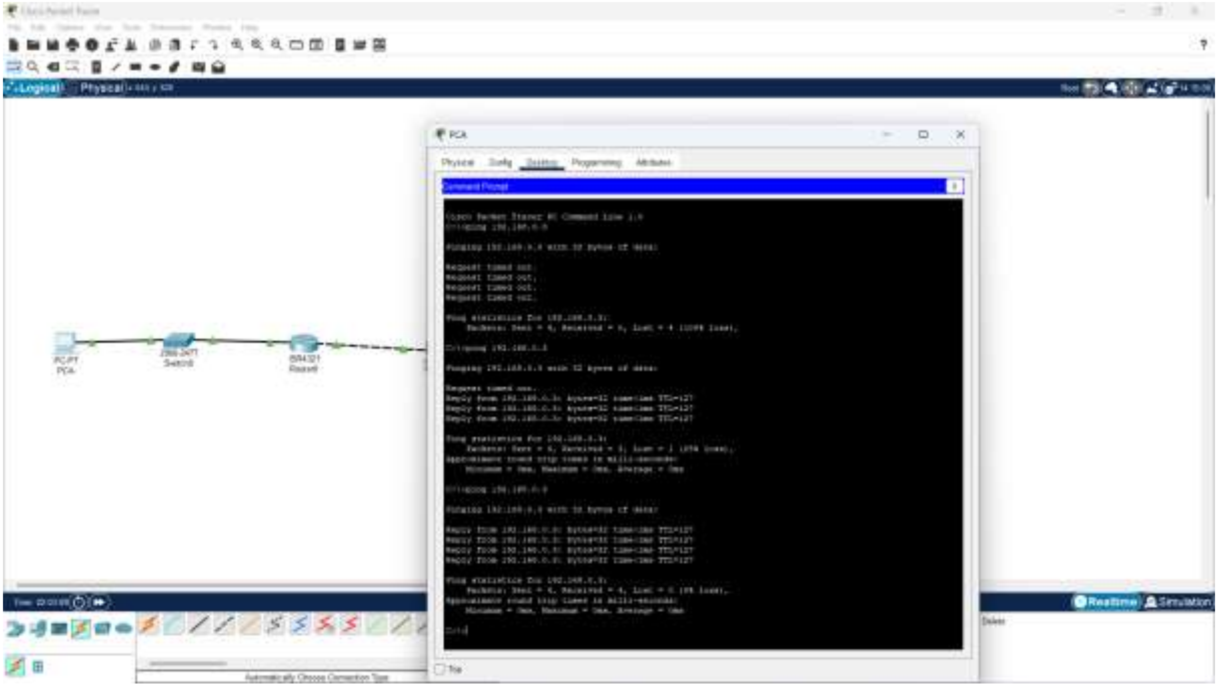
S1(config-if)# **exit**

g. Save the running configuration to the startup configuration file



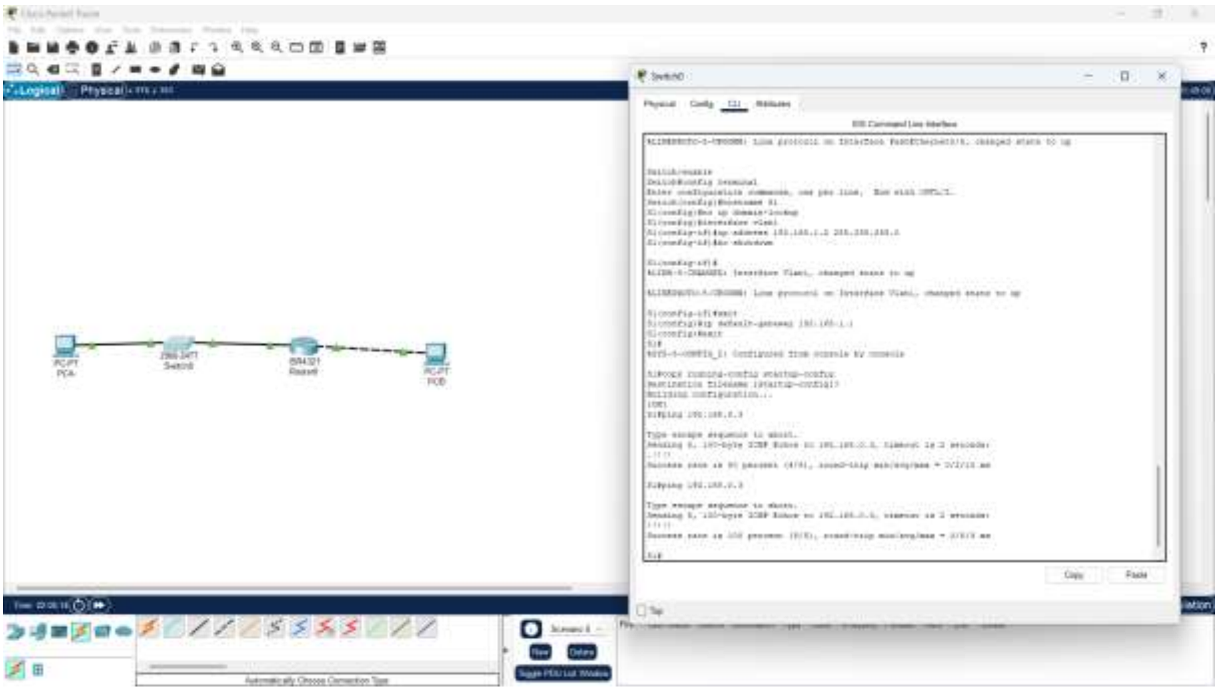
Step 4: Verifying connectivity: end-to-end connectivity.

a. From PC-A, I pinged PC-B to check whether they are communicating



The ping was successful, as illustrated.

b. From S1, I pinged PC-B to check connectivity. I did this two times as shown below. The first attempts gave 80 percent success while the second gave a 100 percent success.



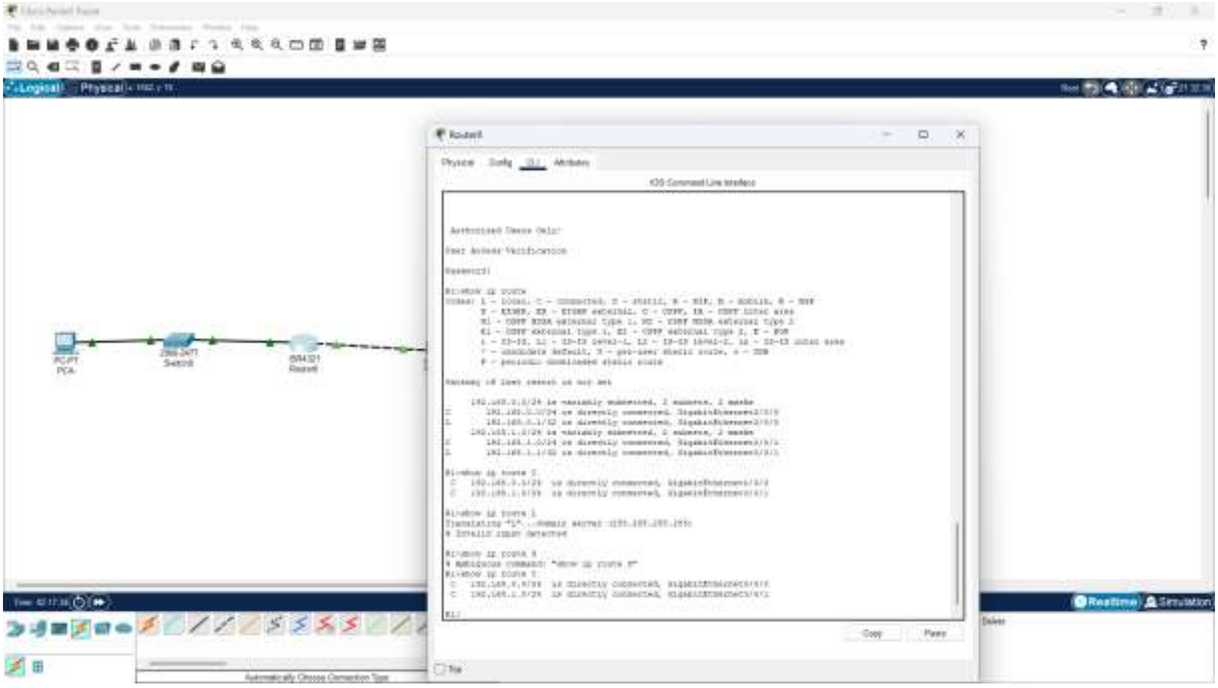
This means the connection was successful.

Part 3: Display Device Information

In this part I used **show** commands to retrieve interface and routing information from the router and switch.

Step 1: Display the routing table on the router.

a. Use the **show ip route** command on the router R1 to answer the following questions



QUESTIONS:

What code is used in the routing table to indicate a directly connected network?

C designates a directly connected subnet. An L designates a local interface.

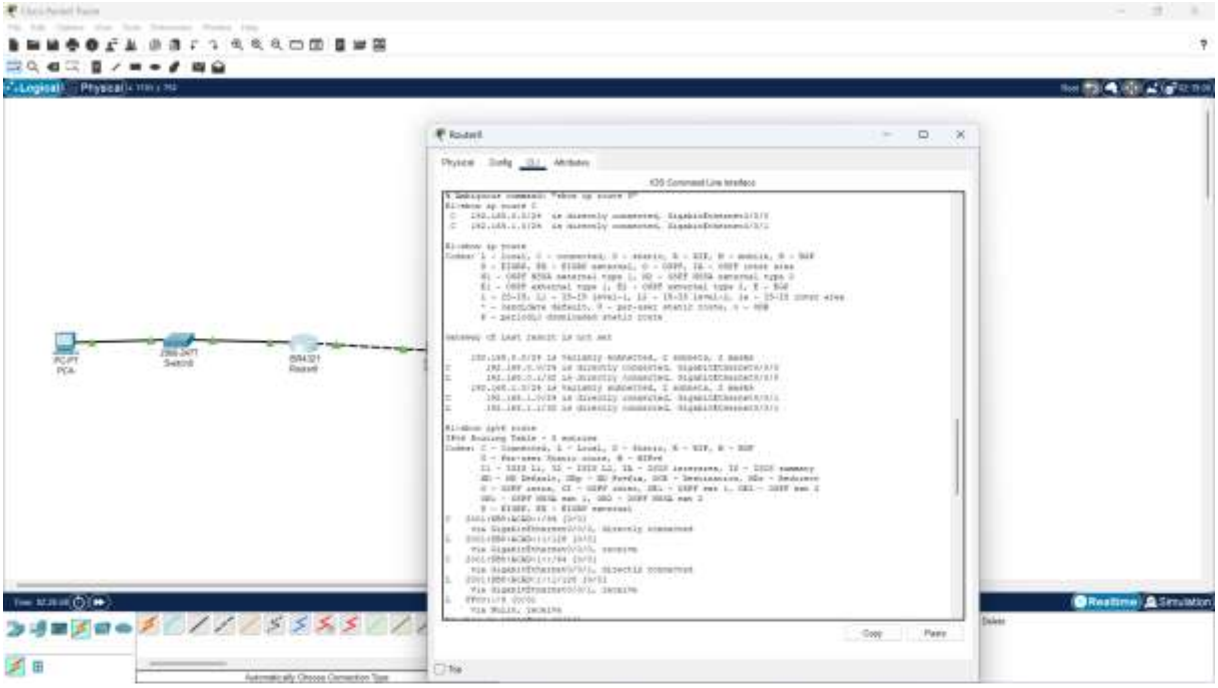
How many route entries are coded with a C code in the routing table?

2

What interface types are associated to the C coded routes?

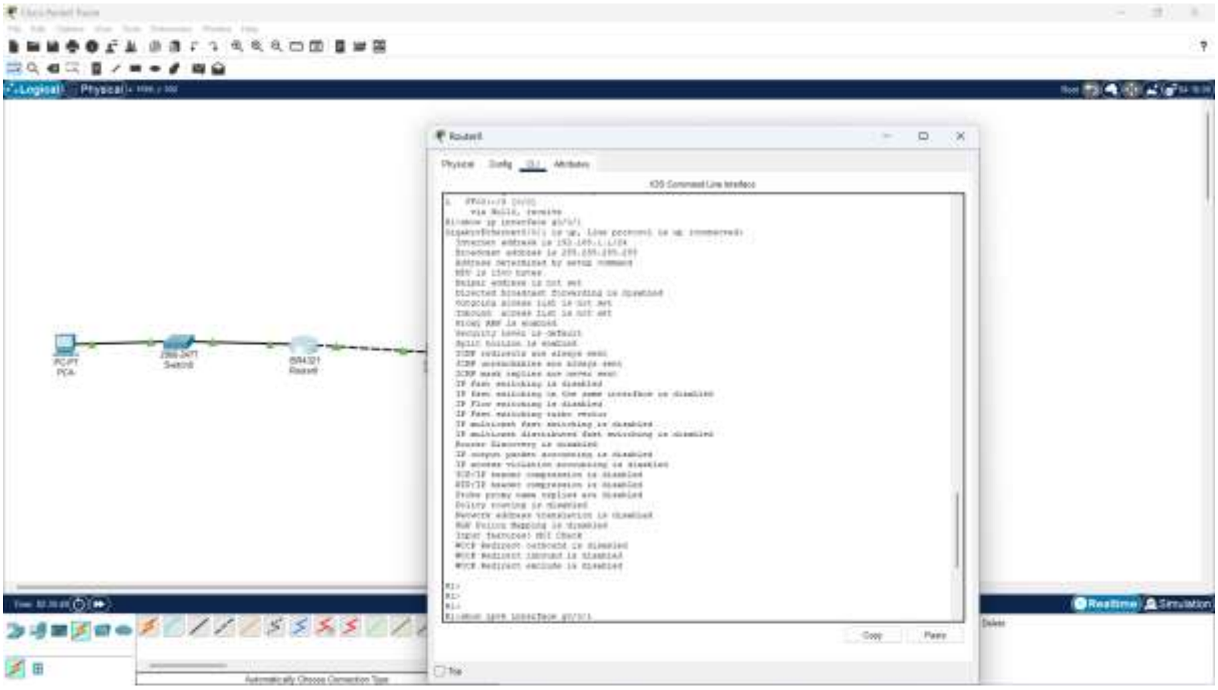
G0/0/0 and G0/0/1.

c.Use the **show ipv6 route** command on router R1 to display the IPv6 routes.



Step 2: Display interface information on the router R1

Use the **show ip interface g0/0/1** to answer the following questions



QUESTIONS

What is the operational status of the G0/0/1 interface?

GigabitEthernet0/0/1 is up, line protocol is up (connected)

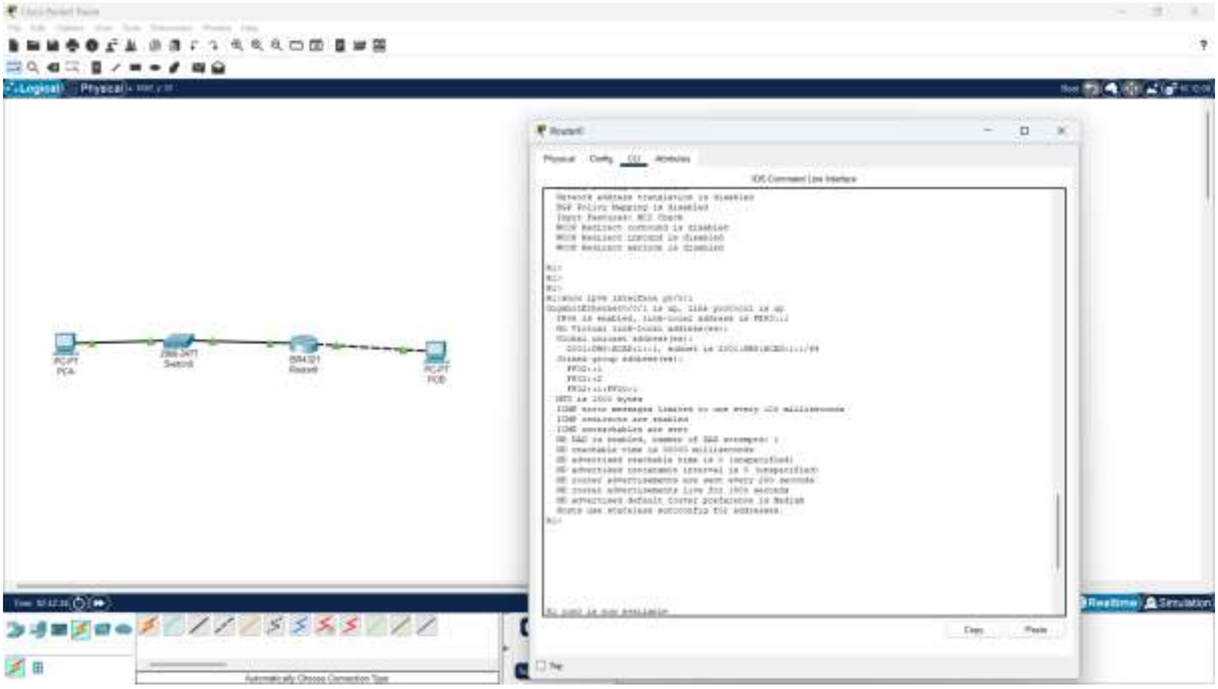
What is the Media Access Control (MAC) address of the G0/0/1 interface?

Hardware is Lance, address is 00d0.5841.8202 (bia 00d0.5841.8202)

How is the Internet address displayed in this command?

Internet address is 192.168.1.1/24

b. For the IPv6 information, enter the **show ipv6 interface *interface*** command. R1# **show ipv6 interface g0/0/1**

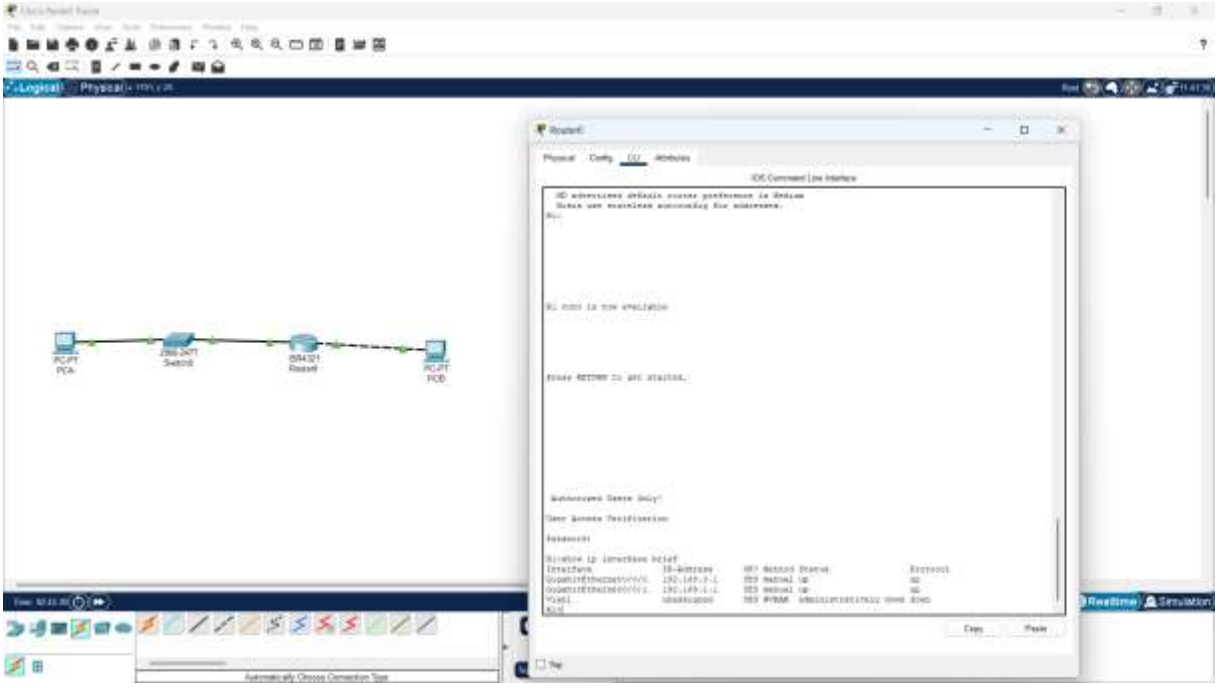


Step 3: Display a summary list of the interfaces on the router and switch.

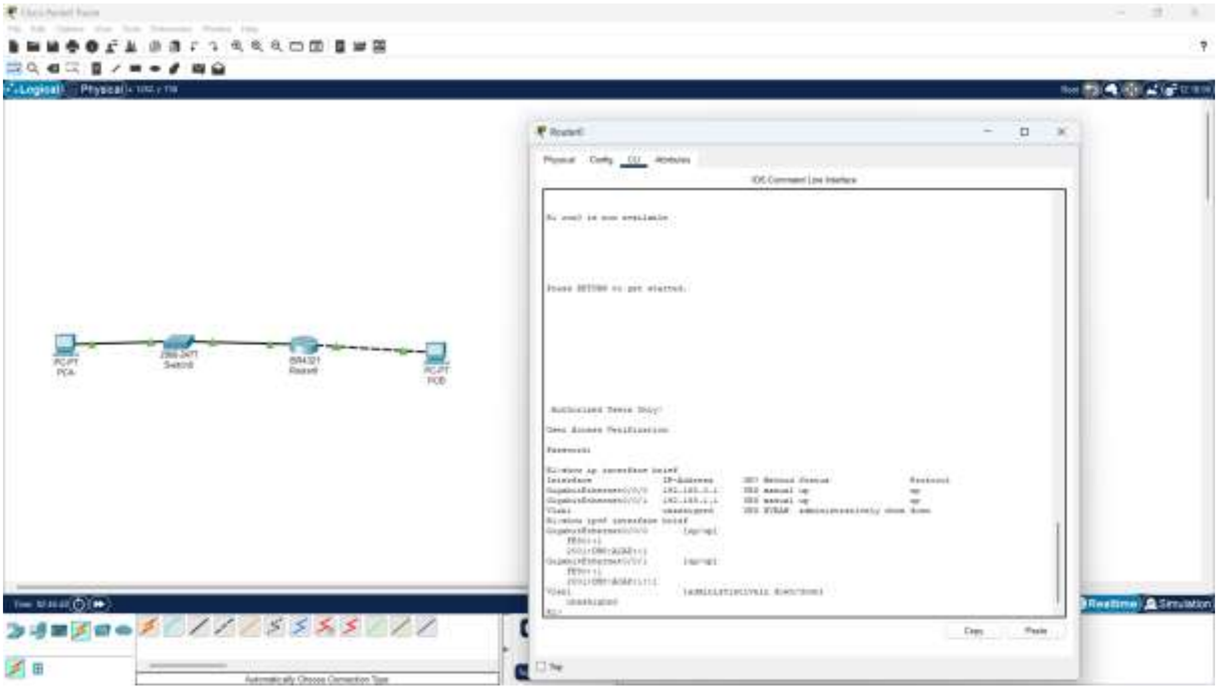
There are several commands that can be used to verify an interface configuration. One of the most useful of these is the **show ip interface brief** command. The command output displays a summary list of the interfaces on the device and provides immediate feedback to the status of each interface.

a. Enter the **show ip interface brief** command on the router R1.

R1# **show ip interface brief**

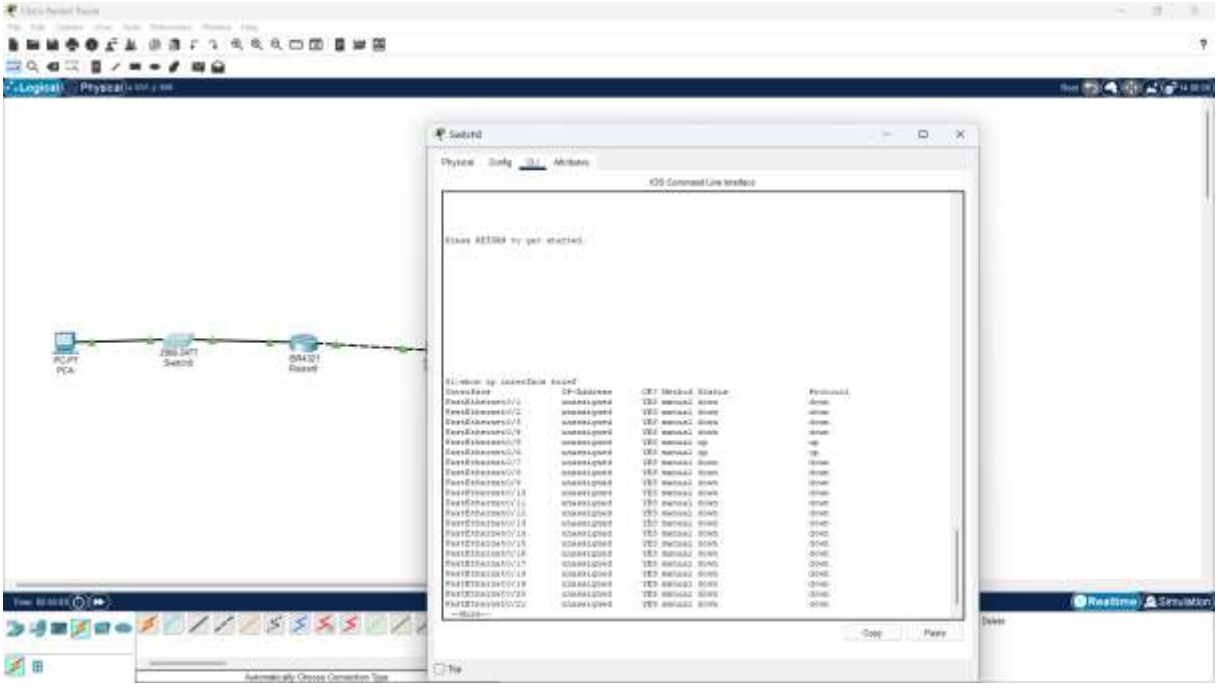


b. To see the IPv6 interface information, enter the **show ipv6 interface brief** command on R1. R1# **show ipv6 interface brief**



c. Enter the show ip interface brief command on the switch S1.

Open configuration windowS1# show ip interface brief



Reflection Questions

1. If the G0/0/1 interface showed that it was administratively down, what interface configuration command would you use to turn the interface up?

    R1(config-if)# no shutdown

2. What would happen if you had incorrectly configured interface G0/0/1 on the router with an IP address of 192.168.1.2?

PC-A would not be able to ping PC-B. This is because PC-B is on a different network than PC-A which requires the default-gateway router to route these packets. PC-A is configured to use the IP address of 192.168.1.1 for the default-gateway router, but this address is not assigned to any device on the LAN. Any packets that need to be sent to the default-gateway for routing will never reach their destination**.**

Conclusion

Completing this lab provided hands-on experience in setting up and configuring a basic switched and routed network. I successfully cabled the devices, configured IP settings, enabled IPv6 routing, secured device access, and verified network connectivity through testing and command-line diagnostics. The activity not only strengthened my practical skills in using Cisco IOS but also enhanced my understanding of how routers and switches operate together within a LAN. This foundational knowledge is essential as I progress toward more complex network scenarios.